

Abstract

Optimal Mixing of Passive Scalars via the Lin–Thiffeault–Doering Velocity: A Pseudo-Spectral Python Implementation

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This thesis investigates the optimal mixing of passive scalars in incompressible fluid flow using the instantaneous-optimal (greedy) velocity field derived by Lin, Thiffeault & Doering (2011). The central objective is to numerically characterise the decay rate of the H^{-1} mix norm under the LTD optimal stirring strategy and to examine how the mixing timescale depends on the spatial scale of the initial scalar distribution.

A full Python implementation of the pseudo-spectral simulation is developed, porting an existing MATLAB codebase to an open, reproducible framework. The scalar transport equation is solved on the two-dimensional torus $[0, 1]^2$ using fast Fourier transforms for spatial discretisation and an adaptive Dormand–Prince RK45 scheme for time integration. Four families of initial conditions — parameterised by a support-size parameter a — are studied, with L^p norm conservation used as a resolution diagnostic and stopping criterion.

Numerical results confirm that the H^{-1} mix norm decays approximately exponentially under LTD stirring, consistent with theoretical predictions. For a representative case ($a = 0.5$), the fitted decay rate is $r \approx 0.44$ (mixing timescale $\tau = 1/r \approx 2.27$). Across eight values of $a \in [0.5, 0.9375]$, the decay rate scales as $r \propto a^{-1.78}$, appreciably steeper than the a^{-1} scaling characteristic of the Iyer–Kiselev–Xu enstrophy-constrained lower bound. A two-resolution comparison ($N = 32$: exponent ≈ -1.61 ; $N = 64$: exponent ≈ -1.78) shows this fitted exponent is resolution-dependent and not yet numerically converged; since the higher-resolution estimate moves farther from, not closer to, the benchmark -1 scaling, these two resolutions alone cannot establish whether the discrepancy reflects a finite-resolution artefact, a genuine feature of the greedy strategy, or both.

The complete Python codebase, Jupyter notebooks, and automated figure generation pipeline are contributed as reproducible research artefacts, providing a foundation for future studies at higher spectral resolution and for comparison with non-greedy global optimisation strategies.

Keywords: passive scalar mixing, H^{-1} mix norm, pseudo-spectral methods, optimal stirring, Lin–Thiffeault–Doering velocity, incompressible fluid mechanics